

E-Commerce Business Insights Report

RFM, Revenue Leakage, Retention, Delivery & Seller Performance Analysis

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Key Customer Segments

Running an RFM analysis on the customer base, looking at how recently people bought, how often, and how much they spent, sorted everyone into five groups: Champions, Loyal, Potential Loyalists, At Risk, and Lost.

One thing jumps out right away: spending a lot doesn't automatically keep someone engaged. There's a customer in the data who spent 772.50 in total but hadn't ordered in 452 days, so they ended up in the Lost category despite being a genuinely high-value buyer. That's a good reminder that "Lost" doesn't always mean "not worth it." Sometimes it just means nobody's followed up in a while.

Major Revenue Risks

The revenue leakage check was built to compare what an order should have cost against what was actually paid, flagging the cases where those two numbers don't match. A number of mismatches did turn up in the process, which points to payment records that aren't fully reconciling with order values.

That's worth taking seriously. Payment and order values need to be monitored more closely going forward so any real leakage or transaction inconsistencies get caught early rather than piling up unnoticed.

Retention Insights

Looking at customers by cohort, grouped by the month of their first purchase, shows that people do come back, just not in a predictable pattern. The February 2017 cohort is a nice example: those customers returned in May, June, August, and November of that year, and then again in January 2018, almost a full year after their first order.

So repeat buying is definitely happening. It's just inconsistent. Some cohorts hang around, others fade fast, and figuring out why is probably the more interesting question here.

Delivery Performance and Customer Reviews

Tracing the order journey: 35,071 orders were placed, 15,312 of those were paid, and 405 made it to the shipped stage. Separately, 34,029 orders across the full dataset were marked as delivered, and 115 came with a review attached in the output reviewed.

As for whether slow delivery actually upsets people, yes, a bit. The correlation between delivery delay and review score is -0.2611, with a slope of -0.03451, so scores do dip as delays stretch out. But the R^2 is only 0.0682, meaning delivery delay only explains about 6.82% of why review scores go up or down. In other words, it matters, but it's far from the whole story. Things like product quality and service are almost certainly doing more of the work.

Seller Performance

Sellers aren't created equal, and the profitability numbers make that obvious. Among the top 10, revenue ranged from around 72,901 to 133,631, but profit margins swung even more, from 72.45% up to 96.39%.

That gap matters. A seller pulling in less revenue can still be the more profitable one to work with. Ranking sellers purely by how much they sell misses that entirely.

What We'd Recommend

- Go after the Lost customers who used to spend well. They're often easier to win back than to replace.
- Take a closer look at the payment-to-order mismatches to pin down how much leakage is actually happening and where it's coming from.
- Study the cohorts that keep coming back and figure out what's different about them.
- Chip away at delivery delays. It won't fix everything, but it'll help review scores.
- Judge sellers on margin, not just revenue.
- Keep an eye on CLV, AOV, Repeat Purchase Rate, and On-time Delivery % on a regular cadence using the Excel KPI tracker.

Overall Executive Insight

Put it all together, and four themes stand out: customer retention, payment accuracy, delivery performance, and seller profitability. The RFM work shows exactly which valuable customers need winning back, the revenue leakage check flags payment issues worth digging into further, the cohort data shows repeat buying is real but uneven, and the stats confirm delivery delays hurt reviews, just not by as much as you might expect. Taken together, this is a solid starting point for tightening up customer management, keeping a closer eye on operations, and being fairer about how seller performance gets judged.